

# Phase 7 : Integration & External Access

## 1.Platform Events

### Purpose

Platform Events in Salesforce enable **event-driven architecture** by allowing systems to communicate asynchronously.  
The Project\_Update\_\_e event is used to notify Salesforce and external systems about updates on project progress, due dates, and notes.

### Configuration Details

#### Event Information

- **Label:** Project Update
- **API Name:** Project\_Update\_\_e
- **Deployment Status:** Deployed

#### Custom Fields

Setup

Home

Object Manager

Quick Find

Setup Home

Salesforce Go

Invoice Setup Assistant

Commerce Setup Assistant

Field Service Setup Home (Beta)

Performance Assistant

Release Updates

Salesforce Mobile App

Lightning Usage

Optimizer

Sales Cloud Everywhere

ADMINISTRATION

Users

Data

Email

LATFOM TOOLS

Subscription Management

Apps

Platform Events

Project Update

Standard Fields (4)

Custom Fields & Relationships (1)

Platform Event Definition Detail

Singular Label

Project Update

Plural Label

Project Updates

Object Name

Project\_Update

API Name

Project\_Update\_\_e

Event Type

High Volume

Publish Behavior

Publish Immediately

Created By

Eshward Santosh Lado

Created On

28/09/2025, 6:23 pm

Modified By

Eshward Santosh Lado

Modified On

28/09/2025, 6:23 pm

Description

Deployment Status

Deployed

Standard Fields

| Action | Field Label  | Field Name  | Data Type       | Controlling Field | Indexed |
|--------|--------------|-------------|-----------------|-------------------|---------|
|        | Created By   | CreatedBy   | Lookup(User)    |                   |         |
|        | Created Date | CreatedDate | Date/Time       |                   |         |
|        | Event UUID   | EventUuid   | Text(36)        |                   |         |
|        | Replay ID    | ReplayId    | External Lookup |                   |         |

Custom Fields & Relationships

New

| Action                                     | Field Label | API Name     | Data Type             | Indexed | Controlling Field | Modified By                               |
|--------------------------------------------|-------------|--------------|-----------------------|---------|-------------------|-------------------------------------------|
| <a href="#">Edit</a>   <a href="#">Del</a> | Date Date   | Date_Date__c | Date                  |         |                   | Eshward Santosh Lado: 28/09/2025, 6:27 pm |
| <a href="#">Edit</a>   <a href="#">Del</a> | Notes       | Notes__c     | Long Text Area(32768) |         |                   | Eshward Santosh Lado: 28/09/2025, 6:28 pm |

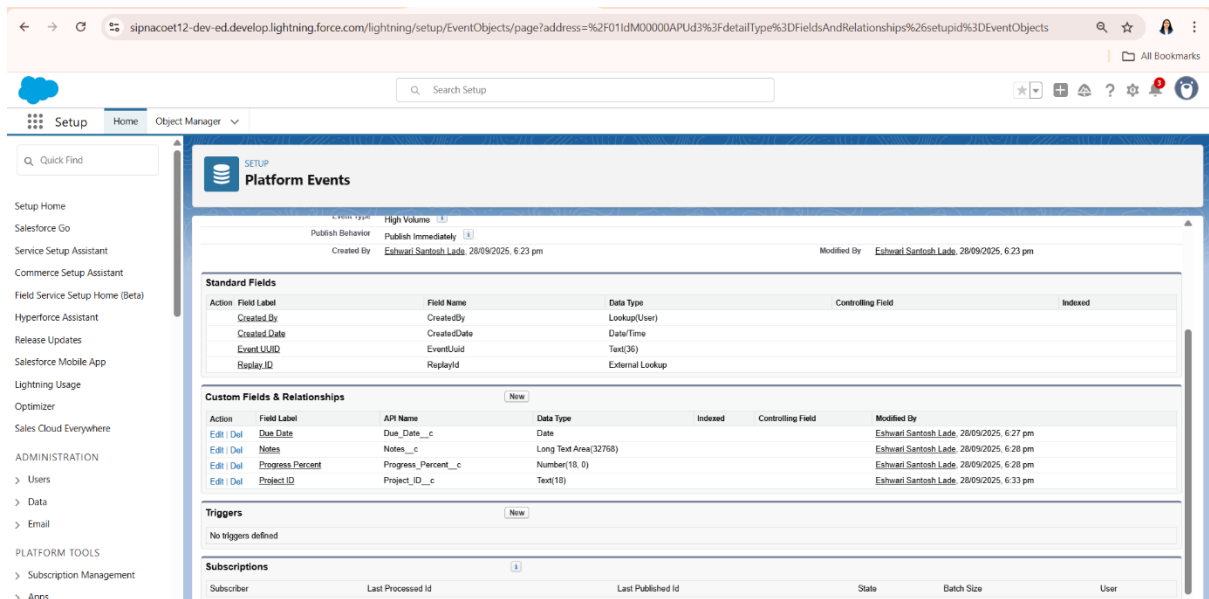


Fig.7.1. Custom Fields Page

| Field Label             | API Name            | Data Type              | Purpose                                               |
|-------------------------|---------------------|------------------------|-------------------------------------------------------|
| <b>Due Date</b>         | Due_Date__c         | Date                   | Tracks expected completion date.                      |
| <b>Notes</b>            | Notes__c            | Long Text Area (32768) | Captures update details or comments.                  |
| <b>Progress Percent</b> | Progress_Percent__c | Number(18,0)           | Records completion percentage of the project.         |
| <b>Project ID</b>       | Project_ID__c       | Text(18)               | Links the event with a Sustainability Project record. |

## Trigger Implementation

### Trigger Name: ProjectUpdateTrigger

- **API Version:** 64.0
- **Status:** Active
- **Purpose:**  
Executes logic after a Project\_Update\_\_e event is published. Currently, it captures and logs the event details. Future enhancements can update related project records or notify stakeholders.

## Code Implementation

```
trigger ProjectUpdateTrigger on Project_Update__e (after insert) {
    for (Project_Update__e ev : Trigger.New) {
        System.debug('Received Project Update: ' + ev.Notes__c);

        // Example placeholder: Update related project record in future
        // if (ev.Project_ID__c != null) {
        //     // Query and update SustainabilityProject__c
```

```
// }  
}  
}
```

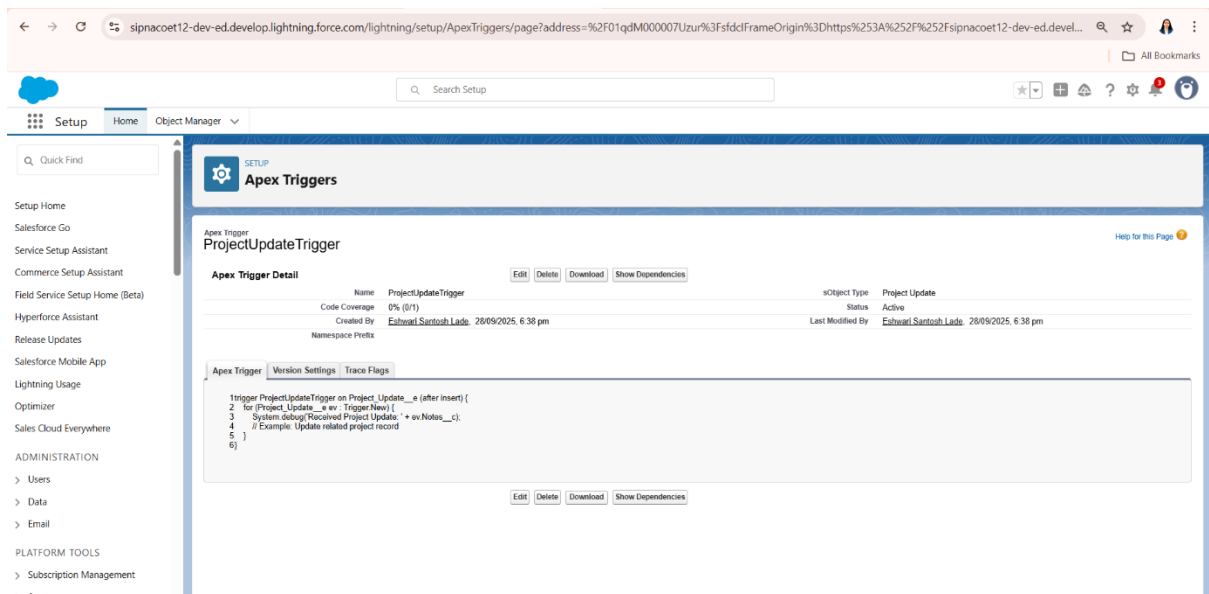


Fig.7.2. Trigger code

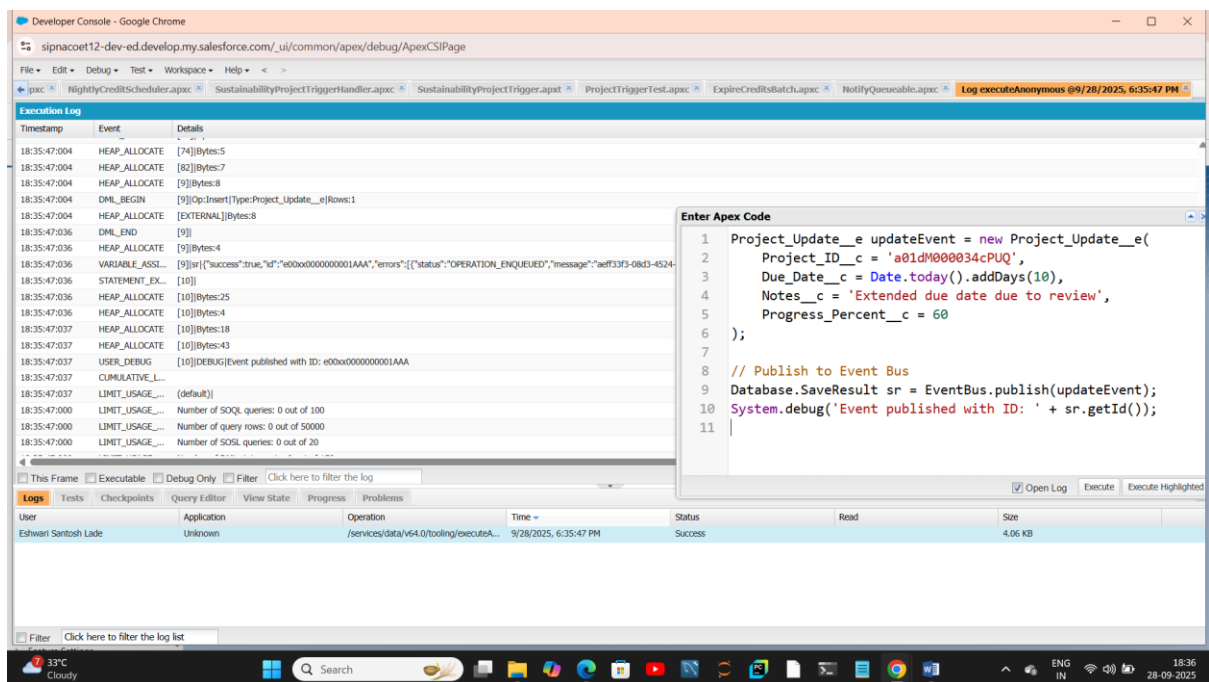


Fig.7.3. Debug Log

## Publishing an Event

Events can be published using **Apex**, **Flows**, or **Process Builder**. Here's an Apex example:

```
Project_Update__e updateEvent = new Project_Update__e(
    Project_ID__c = 'a01dM000034cPUQ',
    Due_Date__c = Date.today().addDays(5),
    Notes__c = 'Phase 1 review completed',
    Progress_Percent__c = 80
);
```

```
Database.SaveResult sr = EventBus.publish(updateEvent);
System.debug('Event published with ID: ' + sr.getId());
```

## Subscriptions

- **Subscriber:** ProjectUpdateTrigger
- **State:** Running
- **Batch Size:** 2000
- **User:** Automated Process

This ensures the trigger listens and processes all incoming events.

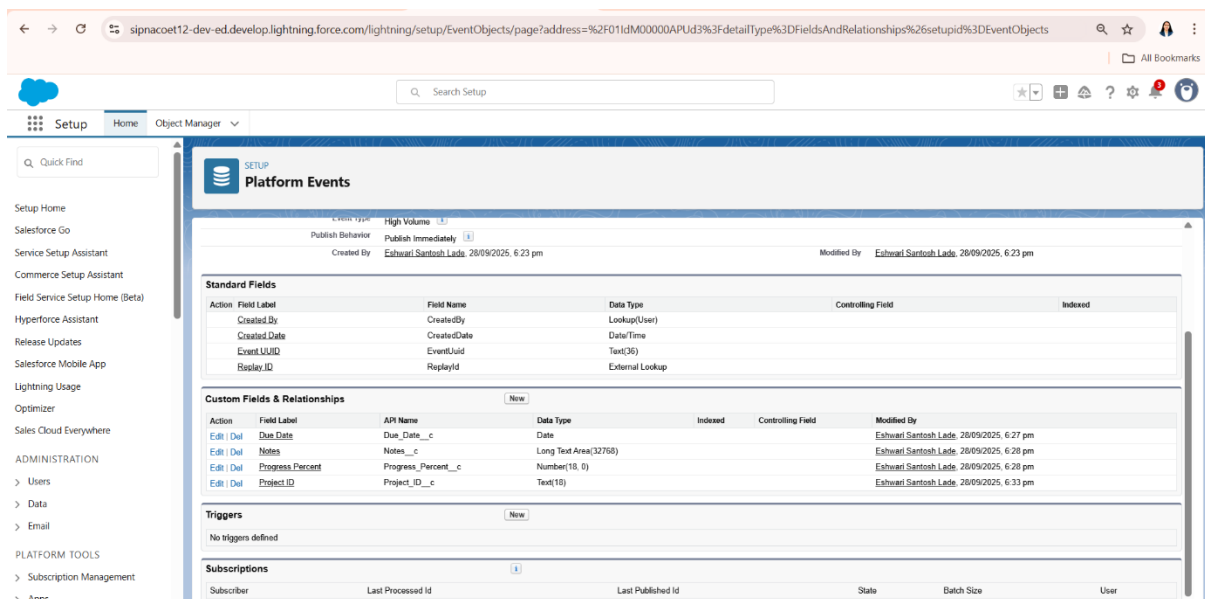


Fig.7.4. Triggers and Subscriptions

## Key Benefits

- Real-time project tracking and updates.
- Asynchronous notifications to Salesforce users and external systems.
- Flexible — can be extended with Flows or Apex.
- Scalable for large event volumes.

## 2.Change Data Capture (CDC)

### Purpose

Change Data Capture (CDC) allows Salesforce to automatically publish **record change events** (create, update, delete, undelete) in **real time**.

It helps synchronize **Salesforce data** with **external systems**, ensuring data consistency across platforms.

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### Configuration Steps

#### 1. Enable Change Data Capture

1. Go to **Setup** → **Change Data Capture**.
2. In the list of available objects, **select the objects** you want to track.
  - Example:
    - Carbon\_Credit\_\_c → to track credit issuance and updates.
    - SustainabilityProject\_\_c → to track project lifecycle changes.
3. Click **Save**.

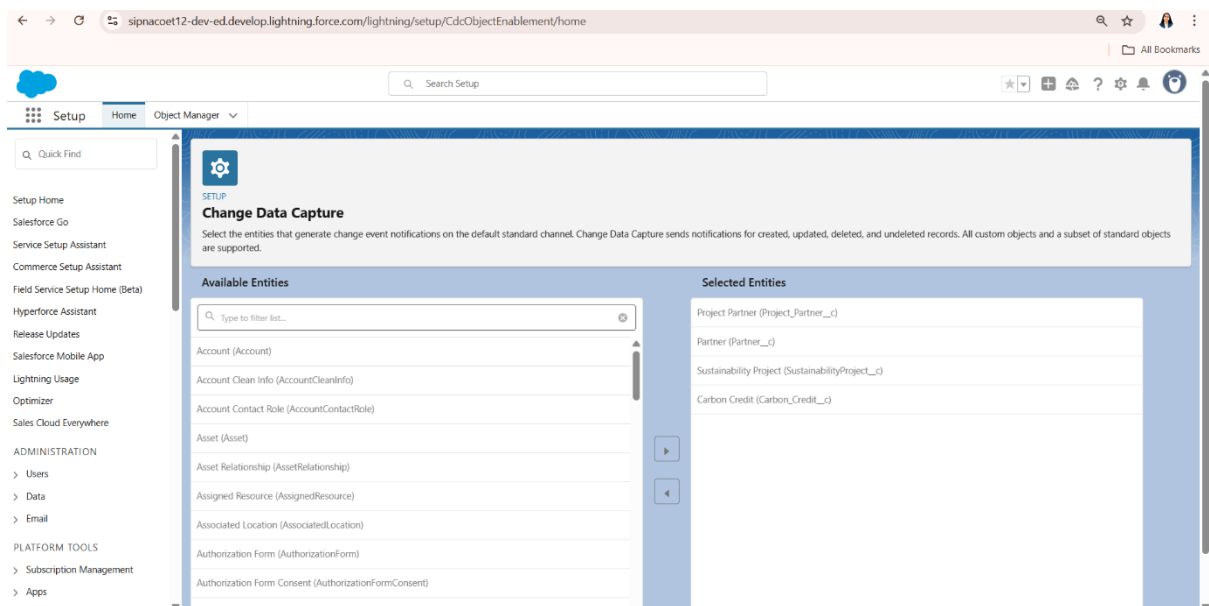


Fig.7.5. Change Data Capture (CDC)

## 3.Salesforce Connect

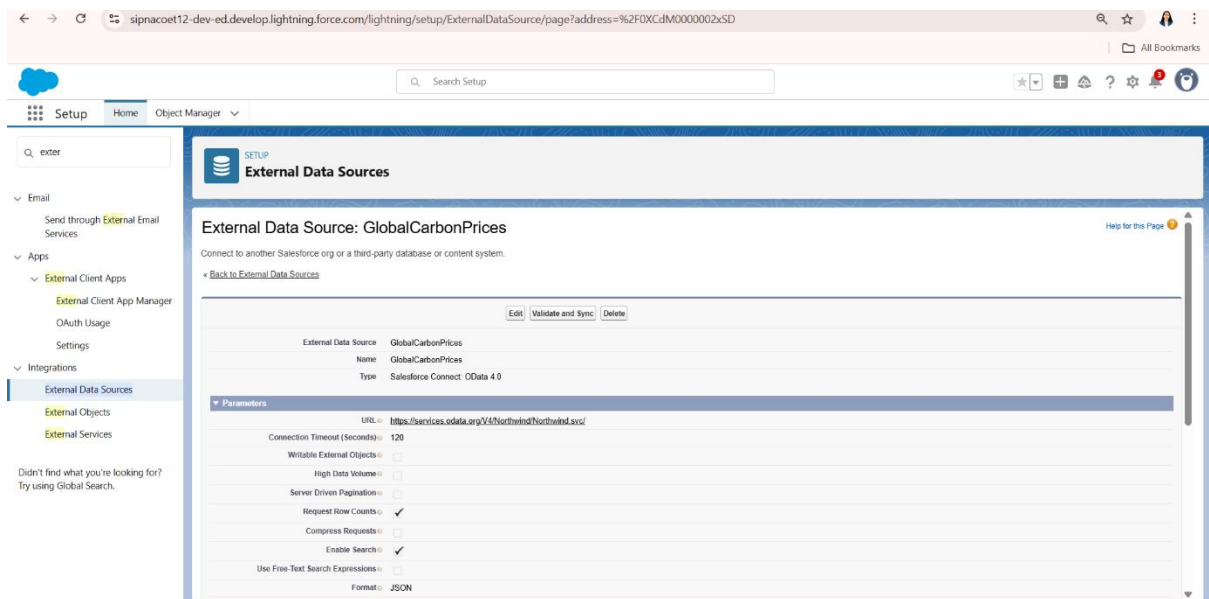
### Purpose:

Enable real-time access to external data within Salesforce without storing it internally. Supports external OData sources for reporting, dashboards, and integrated processes.

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## Steps to Configure:

1. **Navigate to Setup → External Data Sources → New External Data Source.**
2. **Fill in details:**
  - **External Data Source Name:** GlobalCarbonPrices
  - **Type:** Salesforce Connect – OData 4.0
  - **URL:** <https://services.odata.org/V4/Northwind/Northwind.svc/>
  - **Connection Timeout (Seconds):** 120
  - **Format:** JSON
3. **Authentication Settings:**
  - **Identity Type:** Named Principal
  - **Authentication Protocol:** No Authentication (for public OData)
4. **Advanced Options (optional):**
  - Enable or disable server-driven pagination, search, and change data capture.
5. **Click Validate → Sync.**
  - Salesforce creates **External Objects** (e.g., Product\_\_x, Current\_Product\_List\_\_x).
6. **Access External Objects:**
  - App Launcher → Select External Object → View records like normal Salesforce objects.



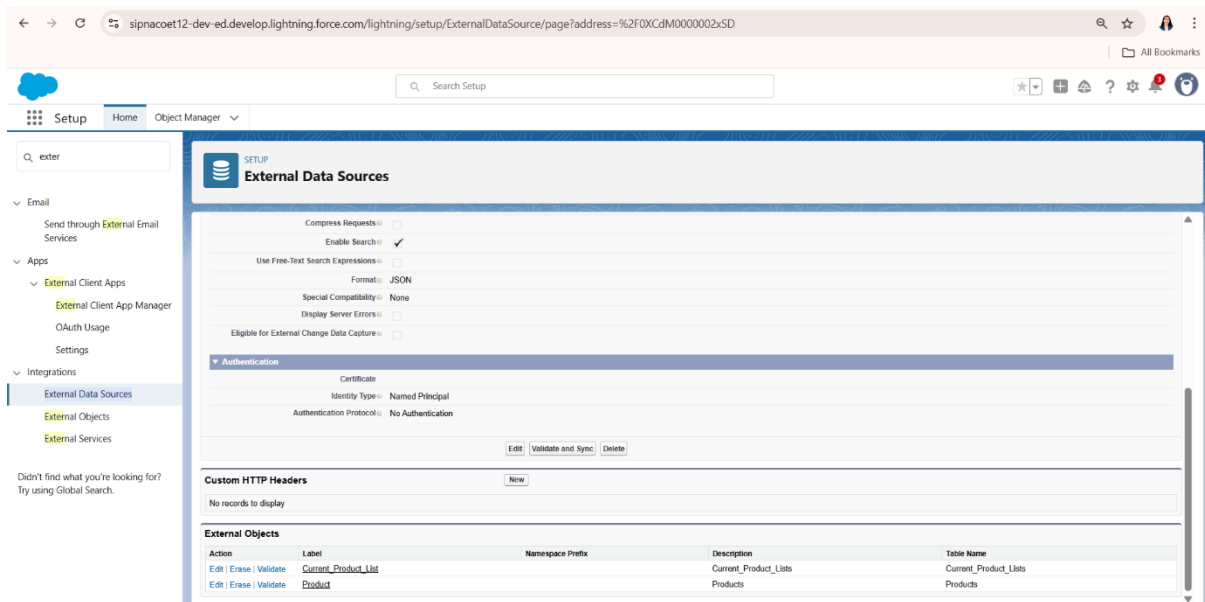


Fig.7.6. Salesforce Connect Page

### Notes / Important Considerations:

- External data is **read in real time**; not stored in Salesforce.
- **External Objects behave like Salesforce objects**: you can include them in reports and dashboards.
- Relationships with internal objects are possible via **External Lookup Relationships**.
- Workflow rules, triggers, or validation rules **cannot** be applied directly to external objects.
- Performance depends on external system response times.

## 4.System Overview

### Purpose:

Provides a snapshot of your Salesforce org's usage, including storage, API calls, custom objects, code, and licenses. Helps in monitoring limits and planning org growth.

### Steps to Access:

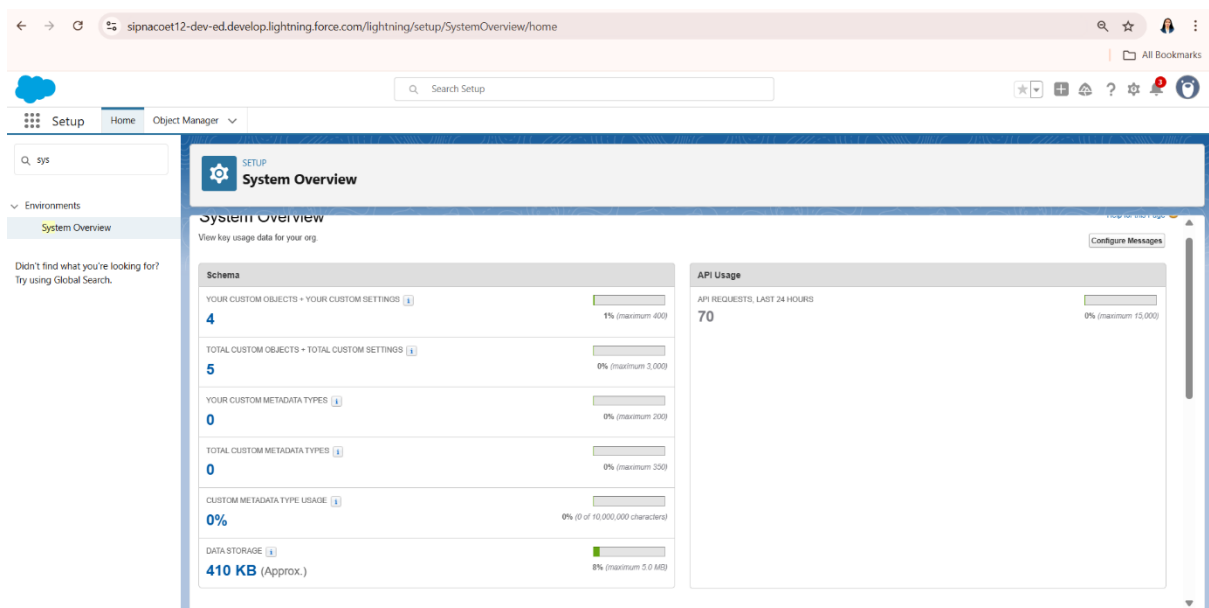
1. Go to **Setup** → **System Overview**.
2. Review key sections:

| Section                      | Details                          | Current Usage | Max Limit |
|------------------------------|----------------------------------|---------------|-----------|
| <b>Schema</b>                | Custom Objects + Custom Settings | 4             | 400 (1%)  |
| <b>Custom Metadata Types</b> | Total & usage                    | 0             | 200 (0%)  |
| <b>Data Storage</b>          | Storage used                     | 410 KB        | 5 MB (8%) |

| Section               | Details                          | Current Usage     | Max Limit                    |
|-----------------------|----------------------------------|-------------------|------------------------------|
| <b>API Usage</b>      | API Requests in last 24 hours    | 70                | 15,000 (0%)                  |
| <b>Business Logic</b> | Rules                            | 11                | 2,000 (0%)                   |
|                       | Apex Triggers                    | 2                 | N/A                          |
|                       | Apex Classes                     | 35                | N/A                          |
|                       | Code Used                        | 22,819 characters | 6,000,000 characters (0.38%) |
| <b>User Interface</b> | Custom Apps                      | 1                 | 10 (10%)                     |
|                       | Custom Tabs                      | 4                 | 100 (4%)                     |
|                       | Visualforce Pages                | 23                | N/A                          |
| <b>Licenses</b>       | Analytics Cloud Integration User | 2                 | 2 (100%)                     |
|                       | Salesforce Platform              | 3                 | 3 (100%)                     |
|                       | Salesforce                       | 2                 | 2 (100%)                     |
| <b>Portal Roles</b>   | Roles                            | 0                 | 50,000 (0%)                  |

### Notes / Recommendations:

- **Monitor API usage:** Ensure integrations do not exceed daily limits.
- **Watch license utilization:** Analytics, Platform, and Salesforce licenses are at 100% usage; request additional licenses if needed.
- **Storage usage:** Data storage is low (8% used), so there's room for growth.
- **Business logic limits:** Rules, triggers, and flows are within safe limits.
- **Custom objects & metadata:** Minimal usage; consider adding new objects carefully to avoid hitting limits.



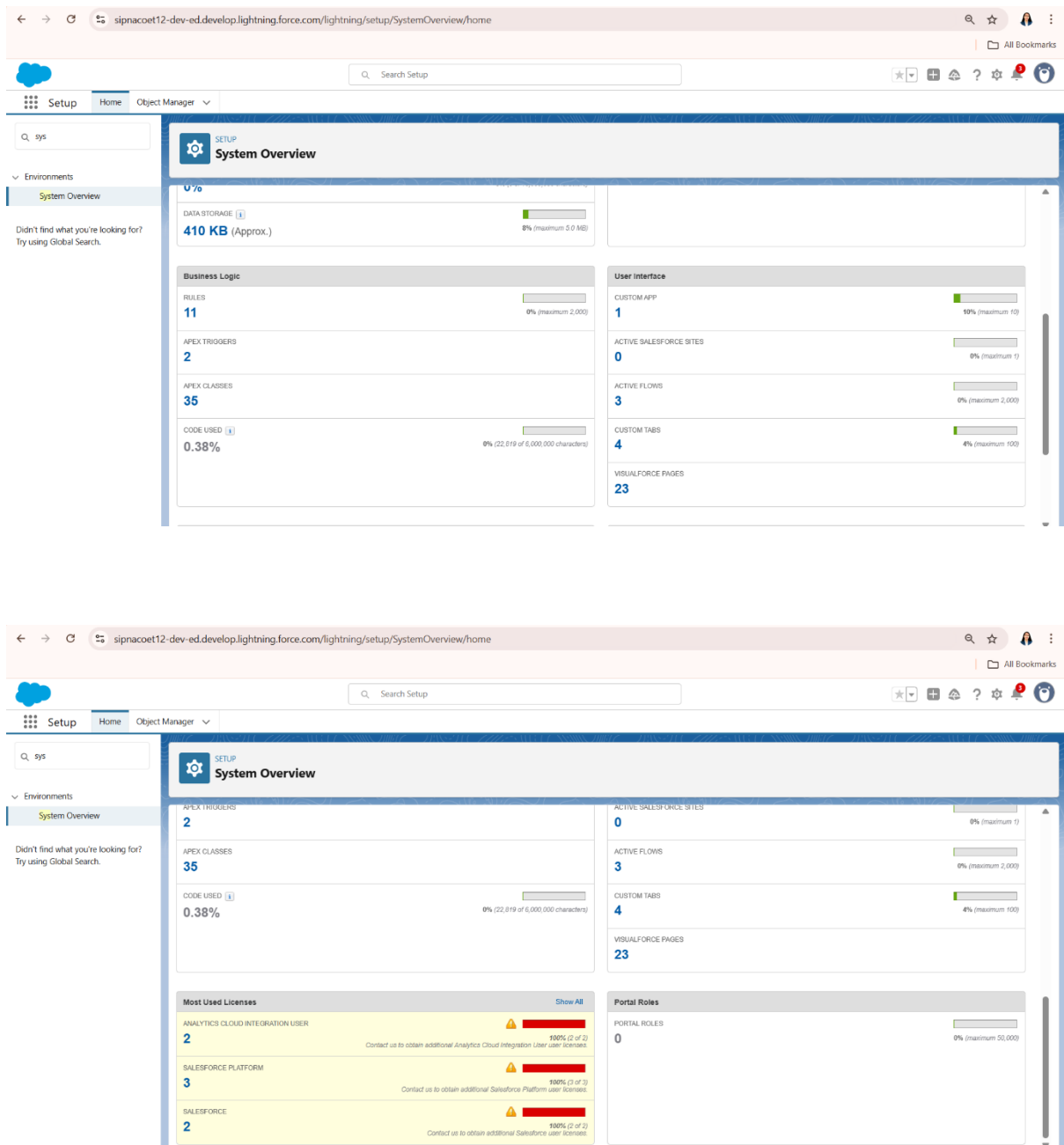


Fig.7.7. System Overview Page

## 5.Remote Site Settings

### Purpose:

The Remote Site Settings configuration allows Salesforce to securely communicate with external web services. In this case, it enables HTTP callouts to the **Carbon Credit API**.

### Implementation Steps:

1. **Navigate to Remote Site Settings**
  - Go to **Setup** → **Security** → **Remote Site Settings**.
  - Click **New Remote Site**.

## 2. Enter Remote Site Details

| Field                            | Description                                                                                  |
|----------------------------------|----------------------------------------------------------------------------------------------|
| <b>Remote Site Name</b>          | Unique identifier for the remote site configuration in Salesforce.                           |
| <b>Remote Site URL</b>           | The base URL of the external API endpoint to allow callouts.                                 |
| <b>Disable Protocol Security</b> | Keeps standard protocol security enabled for HTTPS communication.                            |
| <b>Description</b>               | Brief explanation of purpose for documentation clarity.                                      |
| <b>Active</b>                    | Ensures the configuration is enabled and available for use by Apex, Flows, and integrations. |

### Save Configuration

- Click **Save** to activate the Remote Site.
- Salesforce will now allow Apex classes, Flows, and other integrations to perform HTTP callouts to the Carbon Credit API.

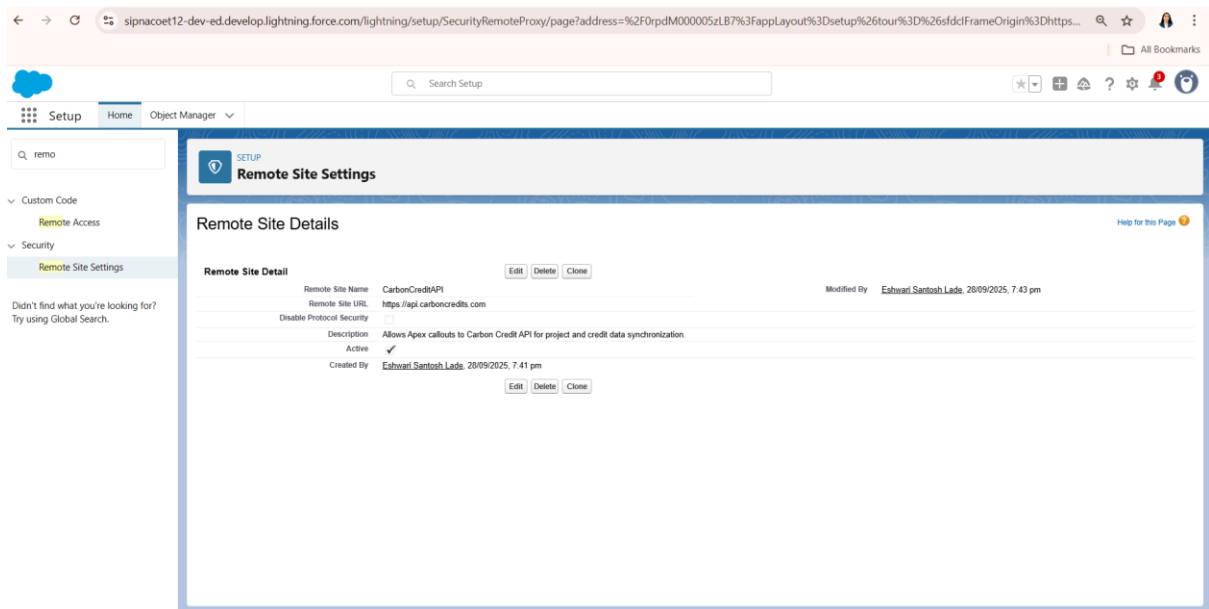


Fig.7.8. Remote Site Settings Page